Conformal Prediction as a Transform within a Grammar for Probabilistic Forecasting

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Abstract

The conformal literature treats rank recalibration as a distinct predictive methodology. We treat it as a univariate transform. Interpolated, the fence-post rank map becomes an invertible operator, exact at its retained levels, within one knot cell everywhere. In a grammar of composable transforms its special status becomes local: it can be used internally, followed by further modeling, pooled, or overruled by the scoring rule. On economic series, refitting through the operator beats stopping at it on three quarters of the series. An exact Bayes pool admits it at bounded cumulative log-loss cost, and a tempered pool at negligible empirical cost.

1 Introduction

In the canonical residual-pooling pattern, a forecaster fits a conditional model and then quotes uncertainty by ranking its past errors with every state pooled. The ranking is honest on average and silent about tonight. That tension, conditional effort upstream and unconditional treatment at the end, is the subject of this paper.

Split conformal prediction can be written as a chain. Fit a model, take residuals, apply the fence-post empirical transform

$$E_n(r) \;=\; \frac{1 + \#\{i \le n : R_i \le r\}}{n+1}, \tag{1}$$

and map the resulting rank back through the model to a prediction set (Vovk et al., 2005; Lei et al., 2018). The value of (1) at a new point is a conformal p-value, and the map itself is the unrandomized conformal predictive distribution of Vovk et al. (2019).

We call it the fence-post transform, after the two +1 corrections: n points cut the line into $n + 1$ cells, and the guarantee rides on the count of cells, not of points. The knot probabilities $i/(n + 1)$ are Weibull's plotting positions, the means of uniform order statistics, so the object is classical.

The literature treats this chain as a construction to be studied whole, and its many variants, normalized, Mondrian, quantile, distributional, and temporal (Boström et al., 2021; Romano et al., 2019; Chernozhukov et al., 2021; Gibbs and Candès, 2021; Auer et al., 2023), as separate constructions. We propose a different organization. The fence-post transform is one operator. The chain is one production rule in a grammar of composable transforms. The interesting question is then not which conformal method to choose but which sequence of transforms minimizes the forecaster's scoring rule.

Stated plainly, conformal prediction is not, in general, an alternative to statistical modeling. It is a particular rank-based transform of a model's output. Its finite-sample property distinguishes it from fitted recalibrators at the point where it is applied. It does not privilege stopping there, it does not remove the modeling that follows, and it does not determine predictive quality under a proper scoring rule.

The standard pattern also carries a conditionality tension. The model predicts conditionally. Its residuals are then pooled across covariate states, and the pooling is justified exactly to the extent that the model has made the residual law conditionally invariant. The certificate does not establish that adequacy. It certifies ranks under exchangeability, so stopping at the pooled law is a modeling decision rather than a consequence of the guarantee. Pooling the observable itself, the limiting case of an inadequate model, would frequently be ludicrous.

In univariate probabilistic forecasting, Platt scaling, isotonic and quantile recalibration, temperature scaling, conformal predictive systems, and distributional conformal prediction can all be represented as monotone recalibration operators on a scalar predictive, score, or probability coordinate, differing in what they fit and what they guarantee (Section 5). The fence-post transform is the member with a certificate, and the certificate is a property annotation, a reason to admit the operator to the grammar, and nothing more.

This reorganization has content because the fence-post operator composes. Once interpolated, E_n is an invertible change of variables with a density, and its probit composition $z = \Phi^{-1}(E_n(r))$ maps exchangeable residuals to approximately standard Gaussian coordinates. Nothing obliges a forecaster to stop there. One can fit further structure on the Gaussianized stream, transform again, and continue, exactly as one does with any other member of the transform library. Conformal calibration stops at the transform. The grammar keeps going.

Two testable claims follow. The first is that the operator has interior value: refitting on the Gaussianized stream can beat stopping at it when exploitable residual structure remains. The second is that nesting is cheap: a pool over chains that includes the conformal pattern should not pay materially for admitting it, because the pool reallocates weight away from the pattern on series where it is a bad choice. We confirm both on 701 economic series and quantify a third finding we did not anticipate: the operator's value depends on the scoring rule for a structural reason, not an implementation one.

A companion paper prices what a fixed representation choice costs (Cotton, 2026b). Under logarithmic loss the irreducible cost of pooling residuals in given coordinates is a mutual information, the conformal information gap, and the companion measures it on the same corpus. The present paper is the constructive complement. If a fixed conformal pattern can be a bad choice with a quantifiable price, the remedy is not a better fixed choice. It is a grammar in which the choice is made by the scoring rule, online, per series.

2 The fence-post transform as an operator

We work with online univariate transforms in the sense of the `skaters` library, described at length in a methods paper (Cotton, 2026d) and briefly in a software paper (Cotton, 2026e). A transform is a pair of functions. The forward function consumes one observation and its own past state and emits a transformed observation. The inverse function maps predictive distributions in the transformed space back to the original space. Conditional on its own past state, the forward map is an invertible change of variables in the current observation, so predictive densities push back exactly, Jacobians included.

The raw fence-post map (1) is a step function. It has no density and no inverse, so it is not a transform in this sense. The minimal repair is interpolation. Let $y_{(1)} \leq \cdots \leq y_{(n)}$ be the past observations, place knots at the order statistics with fence-post probabilities $p_{(i)} = i/(n+1)$ for $i = 1, \ldots, n$, set $z_{(i)} = \Phi^{-1}(p_{(i)})$, and interpolate the pairs $(y_{(i)}, z_{(i)})$ piecewise linearly. We call the resulting operator *gaussianize*. The knot values are the left limits of the right-continuous map (1) at the calibration points, and this convention is what makes the certificate exact at the knots (Proposition 3). That is the full map. The implementation retains at most 39 of the knots, chosen at equal mass from the full sorted history so that each retained knot is an exact order statistic, and drops any knot whose value ties its neighbor, so ties widen the retained cells. We write gaussianize-full for the map through every knot and gaussianize for the implemented retained-knot member. Proposition 5 prices the difference: the certificate is a function of the retained resolution.

The interpolation changes the object, and Section 3 states exactly what survives the change: the discrete map's output is a function of ranks alone, while the interpolated map's output depends on spacings as well, so the two are different operators with different certificates.

Beyond the edge knots the map extends linearly in (y, z) coordinates, and linearity in the probit coordinate is a Gaussian tail in the observable, with scale set by the spacing of the extreme quantiles. Tail mass is unobservable, so any usable version of the operator carries a tail assumption, and this is the mildest one consistent with the coordinates. A kernel-smoothed variant replaces the empirical distribution function with a Gaussian-kernel estimate at Silverman bandwidth, represented by a monotone C^1 cubic through the same knots, and we use it in Section 4.4 to test whether smoothness matters.

Two guards complete the definition, and both were forced by data rather than taste. First, every knot segment's inverse slope dy/dz is capped at twenty times the interquartile inverse slope, so the forward slope is bounded below. An upstream variance collapse can plant a many-sigma outlier as an edge knot, and an uncapped edge segment then hands the inverse map an explosive extrapolation slope. The cap is an explicit assumption: tail scale at most a fixed multiple of the central scale.

Second, the value handed downstream is clamped to $|z| \leq 8$. The clamp applies only to the coordinate supplied to downstream state updates and is not part of the invertible map used to construct or pull back predictive densities, so the change-of-variables identity is intact. When it binds, the downstream learner is updated with a bounded surrogate rather than the literal transformed observation, and that alteration can persist through the learner's state. In the type system the two objects are distinct: gaussianize is invertible and density-carrying, and the bounded update channel is a winsorized learner input, not a transform.

Within the knot range the clamp cannot bind, since no fence-post rank exceeds probability $n/(n+1)$ and $\Phi^{-1}(n/(n+1)) < 8$ for any feasible n. It can bind on the extended tail and on degenerate warmup emissions, and the latter is why it exists, because downstream learners have expanding memory. In one series in our corpus a degenerate warmup emitted a single transformed value near 8000, and the downstream Welford variance and recursive least-squares state never forgot it, inflating that series' loss by four orders of magnitude. Composability is not free. An operator that is harmless as a terminal step can poison downstream learners through one bad emission, and the operator's definition has to prevent this, not the caller.

The guards are better read as declared budgets than as patches. Invertibility alone is too weak a property for a chain. The pullback quantile $y_\alpha = m^{-1}(q(\alpha))$ inherits the steepness of the inverse map m^{-1}, and beyond the edge knots the inverse is an assumption rather than an estimate, because tail mass is unobservable at any n. The slope cap makes the assumption finite. It declares that m^{-1} is Lipschitz, with constant L equal to twenty times the central inverse slope, and the declaration buys a guarantee.

Lemma 1 (Pullback quantiles are budgeted)**.** *Let m be a monotone forward map whose inverse is L-Lipschitz on all of $\mathbb{R}$, extrapolation included, and let q be the quantile function of the predictive in transformed coordinates. Then the pulled-back predictive has quantiles $y_\alpha = m^{-1}(q(\alpha))$ and*

$$|y_\alpha - y_{1/2}| \;\le\; L\left|q(\alpha) - q(\tfrac{1}{2})\right| \qquad \textit{for every } \alpha \in (0,1).$$

For a chain of such maps the same bound holds with L the product of the rungs' constants.

Proof. Monotone maps transport quantiles, so $y_\alpha = m^{-1}(q(\alpha))$. The Lipschitz bound applied to the pair $q(\alpha), q(\frac{1}{2})$ gives the display. A composition of Lipschitz maps is Lipschitz with the product constant. □

With a Gaussian leaf of scale σ the right side is $L\sigma\,|\Phi^{-1}(\alpha)|$, so no realization of the data can make a quoted quantile explode. The naive empirical map declares $L = \infty$ and buys nothing. The clamp is the same discipline in the forward direction: the codomain bound $|z| \le 8$ is intrinsic to the fence-post ranks, and a downstream learner with expanding memory is safe against exactly the emissions the bound excludes. Any finite multiplier yields the lemma, so the value twenty is the one remaining choice of taste, a tail-conservatism dial. Fitting the tail where the sample reaches, generalized-Pareto by censored maximum likelihood as the leaf already does, with the budget kept as a prior bound, is the refinement we leave open.

Before enough observations accumulate, the operator standardizes by expanding moments, which is the $n \to 0$ limit of the empirical map. After the warmup the knots are rebuilt each step, at most thirty-nine of them at equal mass from the full sorted history, so the retained knots sit at exact order statistics and carry no rank-approximation error. The sorted buffer itself grows with the series. A bounded compacted sketch is the obvious refinement, and its price is the rank-error term of Proposition 5.

3 What composition preserves

Each operator in a grammar carries properties. It may be monotone, invertible, density-carrying, bi-Lipschitz with a declared budget, exchangeability-preserving, or finite-sample calibrated. Composition preserves some and destroys others, and the pattern is positional. We record the four facts that organize our experiments.

Proposition 2 (Fence-post uniformity)**.** *Let $R_1, \ldots, R_{n+1}$ be exchangeable and almost surely distinct. Then $E_n(R_{n+1})$ is uniform on $\{1/(n{+}1), \ldots, (n{+}1)/(n{+}1)\}$. In particular,* $\Pr\{E_n(R_{n+1}) \le k/(n+1)\} = k/(n+1)$ *for $k = 1, \ldots, n+1$.*

Proof. By exchangeability every ordering of $R_1, \ldots, R_{n+1}$ is equally likely, and by almost-sure distinctness the rank of R_{n+1} among the $n+1$ values is uniform on $\{1, \ldots, n+1\}$. The map (1) evaluated at R_{n+1} equals the rank divided by $n+1$. □

This is the standard conformal fact (Vovk et al., 2005) stated as a property of the operator's output. It certifies the discrete map (1), and the discrete map is what the experiments do not run. The interpolated operator of Section 2 uses spacings as well as ranks, and its output between knots is not distribution-free. What survives interpolation is the grid.

Proposition 3 (Grid calibration survives interpolation)**.** *Let $Y_1, \ldots, Y_{n+1}$ be exchangeable and almost surely distinct. Let m be any strictly increasing map, interpolation and tails arbitrary, that passes through the knots: $m(y_{(i)}) = z_{(i)}$ at the order statistics $y_{(1)} < \cdots < y_{(n)}$ of $Y_1, \ldots, Y_n$. Then*

$$\Pr\{\, m(Y_{n+1}) \le z_{(i)} \,\} \; = \; \frac{i}{n+1}, \qquad i = 1, \ldots, n.$$

Proof. Monotonicity gives $\{m(Y_{n+1}) \le z_{(i)}\} = \{Y_{n+1} \le y_{(i)}\}$. By exchangeability and distinctness the rank of Y_{n+1} among all $n+1$ values is uniform on $\{1, \ldots, n+1\}$, and $Y_{n+1} \le y_{(i)}$ holds for exactly the i lowest ranks. □

There is no need to insist on exactness off the grid, because the relaxation is uniformly small and assumption-free. Call an operator *δ-conformal* if, on exchangeable almost surely distinct input, its output U satisfies $\sup_u |\Pr\{U \le u\} - u| \le \delta$.

Proposition 4 (Interpolation is $1/(n+1)$-conformal)**.** *In the setting of Proposition 3, write $U = \Phi(m(Y_{n+1}))$ for the probability-scale output of any interpolated map through the knots. Then*

$$\sup_{u \in (0,1)} \big| \Pr\{U \le u\} - u \big| \; \le \; \frac{1}{n+1},$$

whatever the interpolation, the tails, and the continuous input law. Equivalently, $m(Y_{n+1})$ is within Kolmogorov distance $1/(n+1)$ of standard Gaussian.

Proof. For u between consecutive grid levels $i/(n+1)$ and $(i+1)/(n+1)$, monotonicity of the distribution function and Proposition 3 trap $\Pr\{U \le u\}$ in $[\, i/(n+1),\, (i+1)/(n+1) \,]$, the same cell that contains u. Two points of one cell differ by at most $1/(n+1)$. The edge cells are the same argument with endpoints 0 and 1. □

The bound generalizes to the implemented operator, which does not keep every knot.

Proposition 5 (The certificate scales with the retained resolution)**.** *Let $0 = r_0 < r_1 < \cdots < r_K < r_{K+1} = n+1$, and let m be any strictly increasing map with $m(y_{(r_j)}) = \Phi^{-1}\big(r_j/(n+1)\big)$ at the retained order statistics, interpolation and tails arbitrary. Then $U = \Phi(m(Y_{n+1}))$ satisfies*

$$\sup_{u \in (0,1)} \big| \Pr\{U \le u\} - u \big| \le \Delta_n := \max_{0 \le j \le K} \frac{r_{j+1} - r_j}{n+1}.$$

If each retained knot's value sits within e stream ranks of its nominal rank, the bound degrades to $\Delta_n + e/(n+1)$.

Proof. The argument of Proposition 3 gives exactness at each retained level $r_j/(n+1)$. Between consecutive retained levels, monotonicity traps $\Pr\{U \le u\}$ and u in one retained cell, of width at most Δ_n. A rank error of e shifts a retained exact level by at most $e/(n+1)$, which adds to the cell width. □

The full map is the case $\Delta_n = 1/(n+1)$. The implemented operator retains at most 39 knots at exact order statistics, so $e = 0$ and after warmup its certificate is roughly one part in forty everywhere, exact at the retained levels. The e term is the price a future bounded-memory compaction would pay. A dropped tie widens the retained cell that absorbs it, so tie mass enters through Δ_n, and randomizing within tie blocks would restore exactness at the price of determinism. The corpus of Section 4 filters to repeat fraction below five percent rather than zero, so the tie term is small and not vanishing. The statement worth keeping is that the certificate is a function of knot resolution, sketch error, and tie mass: the guarantee is priced in the operator's computational budget.

The discrete map earns the same δ: its output is supported on the grid, so as a distribution on $[0, 1]$ it too sits within Kolmogorov distance $1/(n+1)$ of uniform. At the resolution the sample can support, the two operators are equally conformal, and the interpolated one is invertible.

What separates them is the exact statement. The discrete map is deterministic, exactly calibrated at every grid level, noninvertible, and $1/(n + 1)$-calibrated between grid levels. Randomized smoothing, which spreads each output uniformly over its rank cell, is exactly calibrated at every level and is a stochastic kernel rather than a deterministic transform (Vovk et al., 2005). The interpolated map is deterministic and invertible, stays exact on the grid, and incurs at most one cell of calibration error elsewhere. The grammar uses the deterministic, invertible version and quantifies its one-cell calibration relaxation.

The exact conformal object is a rank certificate, and a rank certificate is not a predictive density. Turning it into one requires interpolation, smoothing, tail assumptions, and model selection, the same choices met everywhere else in probabilistic forecasting, and exchangeability supplies none of them.

The log score will still notice: it integrates the predictive density between grid points and in the tails, exactly where the certificate is δ-relaxed and assumption-laden, and Section 4.4 measures what that costs.

The second fact is that this guarantee is positional. Proposition 2 concerns the operator's immediate output. If a downstream model is fitted on $z_1, \ldots, z_t$ and its parameters enter later forecasts, the composed system's forecast at time $t + 1$ depends on the ordering of the past stream, and the exchangeability that delivered uniformity no longer applies to the composed output. The calibration certificate attaches to a position in the chain, not to the chain. This is not a defect. It is the familiar price of fitting, and the reason the companion paper's ledger separates representation, model-form, and estimation cost (Cotton, 2026b). But it means the grammar has a typing discipline: an operator's guarantee names the position at which it holds, and claims about the composed system need their own arguments.

The third fact concerns schedules, and it explains what the conformal literature's data splitting is for and how an online chain does without it. Call a chain *previsible* if the map applied to each observation is measurable with respect to information strictly prior to that observation. Dual use, the defect the splitting exists to avoid, is exactly a previsibility violation: an observation flowing through a map whose state already contains it.

Split conformal enforces previsibility with a coarse clock. Interior operators learn on a fitting block and freeze, the terminal fence-post learns on a disjoint calibration block and freezes, and test points meet only frozen maps.

The chains of this paper enforce the same rule with the finest clock. Every operator emits before it absorbs, so the map applied at time t is built from $y_1, \ldots, y_{t-1}$ alone, and the past is a calibration set that never contains the present. Batching is one scheduler for one rule, and updating after emission is another.

On the exchangeable hypothesis the fine clock does not merely match the batch guarantee. It strengthens it.

Proposition 6 (Sequential fence-post outputs are independent)**.** *Let $Y_1, \ldots, Y_T$ be exchangeable and almost surely distinct, and let $U_t = E_{t-1}(Y_t)$ be the fence-post transform of Y_t through the* *map built from $Y_1, \dots, Y_{t-1}$, as in (1) with $n = t-1$. Then $U_1, \dots, U_T$ are independent, and U_t is uniform on $\{1/t, \dots, t/t\}$.*

Proof. Let $K_t \in \{1, \dots, t\}$ be the rank of Y_t among $Y_1, \dots, Y_t$. From (1) with $n = t-1$, $U_t = \big(1 + \#\{i < t : Y_i \le Y_t\}\big)/t = K_t/t$. The full ranking of $Y_1, \dots, Y_T$ is a permutation π of $\{1, \dots, T\}$, and the map $\pi \mapsto (K_1, \dots, K_T)$ is a bijection onto $\{1\} \times \{1,2\} \times \cdots \times \{1, \dots, T\}$: given the sequential ranks, the permutation is reconstructed by inserting each element at its stated position, and both sets have cardinality $T!$. Exchangeability with almost-sure distinctness makes π uniform over the $T!$ orderings, so $(K_1, \dots, K_T)$ is uniform over the product set. Uniformity over a product set is independence of the coordinates, each uniform on its factor. □

The independence of sequential ranks is classical (Rényi, 1962) and powers online exchangeability testing (Vovk et al., 2003). Here it separates the schedules. Split conformal's p-values share one calibration set and are dependent. The prequential outputs are independent, and the price is early resolution: the map at time t has granularity $1/t$, hence the warmup of Section 2.

The certificate survives any further fixed monotone map, the probit included, because monotone maps preserve ranks. It dies at the first downstream fit, by the second fact, and a chain that wants a terminal certificate ends with one more fence-post step on the final residuals. Full conformal is a third schedule, symmetrizing the whole chain over the data at the cost of refitting per candidate value (Vovk et al., 2005). We do not use it. An operator's annotation is therefore a position and a clock.

For the economic series of Section 4 the exchangeable hypothesis of Proposition 6 is false, as it is throughout time-series conformal prediction, so the uniformity claims are conditional on model adequacy. Previsibility is not. It holds by construction, and every score in this paper is out-of-sample even where the guarantee hypothesis fails. The contracts so far assemble into the beginning of the type table the conclusion asks for:

operator	previsible	exch.-preserving	invertible	certificate
fence-post, discrete	yes	independent sequential ranks	no	exact at grid levels
gaussianize, full grid	yes	not claimed	yes	exact at every level, $1/(n{+}1)$ off
gaussianize, implemented	yes	not claimed	yes	exact at 39 retained levels, Δ_n off
fixed monotone link	yes	yes	yes	transparent
EMA location, EWMA scale	yes	no	given state	none
frozen split-fitted model	yes, coarse clock	cal/test pairs	varies	none
refit on the stream	yes	no	given state	none; blocks pass-through

The middle column is the working hypothesis of every chain in Section 4: location and scale do not preserve exchangeability, so the fence-post rung downstream of them operates under an idealization, which the contracts state rather than hide.

The fourth fact prices the choice of coordinates. The companion paper's transform-pooling theorem states that pooling residuals R after an invertible state-dependent change of coordinates ϕ_X has irreducible log-score gap $\mathrm{I}(\phi_X(R); X)$, the mutual information between the transformed residual and the state X on which its true conditional law depends: the Jacobians of the transform cancel, and only the coordinates' ability to remove conditional heterogeneity matters (Cotton, 2026b). The gaussianize operator is a data-dependent monotone map, measurable with respect to its own past state, and piecewise C^1, so conditional on the state the theorem's hypotheses hold. The grammar view adds only this: since operators compose, the coordinates in which one pools are themselves a product of the chain, and the gap is a property of the whole chain up to the pooling point.

The operator's ancestry is long. The branches are not one idea. They transform different objects for different reasons, and what they share is the trunk: push an observation through a distribution function and work on the flat scale.

The empirical distribution function is Galton's ogive (Galton, 1875). Plotting positions place ordered observations at probabilities of the form $i/(n+1)$ (Hazen, 1914), and the probability integral transformation received its name and first systematic treatment from Pearson (1938). Normal scores, the replacement of ranks by standard-normal quantiles, are the gaussianize map's direct ancestor (van der Waerden, 1952). That estimation spoils the transform's exactness has been known since David and Johnson (1948), and the positional fact above is its descendant.

Rosenblatt's chaining through conditional distributions whitens any process (Rosenblatt, 1952), and normalizing flows are its learned, offline descendants (Tabak and Vanden-Eijnden, 2010; Papamakarios et al., 2021). Parametric monotone Gaussianizers run from Box and Cox (1964) to Yeo and Johnson (2000), quantile mapping in the geosciences is the same move under another name (Panofsky and Brier, 1968), transformed residuals as model criticism begin with Cox and Snell (1968), the PIT as a forecast-evaluation instrument with Diebold et al. (1998), and the prequential stance that all of this paper's scoring obeys is Dawid (1984).[1]

The transform also ran as a market before it ran here as an operator. The microprediction platform's z-streams solicited competitive real-time predictions of PIT-transformed residual streams (Cotton, 2022), chaining solicitations so that one crowd predicted the residuals of another's forecasts. That is interior composition as a mechanism, and the algebra of such mechanisms is developed in Cotton (2026a) and Cotton (2026c). The grammar of this paper is the single-forecaster form of the same design.

Against this ancestry the fence-post operator's distinction is narrow and real: at its position, on exchangeable input, its ranks carry a finite-sample certificate.

4 Measurements

4.1 Design

The corpus, scoring conventions, and evaluation protocol follow the companion paper (Cotton, 2026b). We use 701 FRED change series with at least 600 observations and repeat fraction below five percent. Every forecaster runs prequentially: at each time it issues a one-step predictive density from data strictly before that time, then sees the observation and updates. Scoring starts after a burn-in of 300 observations. The predictive in the original coordinates is the finite Gaussian mixture produced by the nine-particle inverse. That implemented mixture, blended with weight 0.01 toward an expanding-window Gaussian reference, is scored exactly, with no clipping or flooring in the reported nats. CRPS is computed in closed form from the Gaussian-mixture predictive. Per-series CRPS is normalized by the location baseline defined next, so scales are comparable across series.

Six single chains and three pools are compared. All chains end in a Gaussian leaf with online variance, and all transforms update online.

[1] A fuller annotated timeline is maintained at `https://skaters.microprediction.org/heritage.html`.

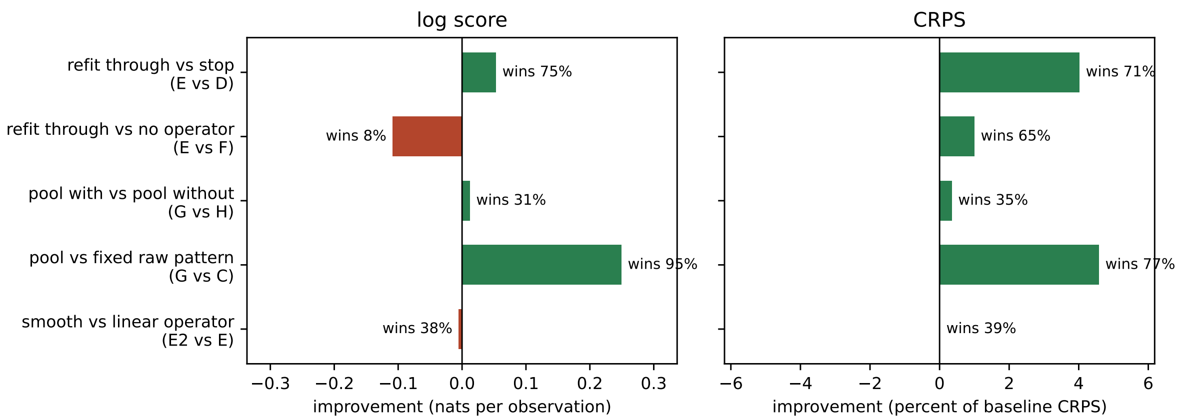

Figure 1: The five comparisons behind the three findings, on 701 series under both scoring rules. Bars to the right favor the first-named config. Log score improvement is in nats per observation. CRPS improvement is in percent of the location-baseline CRPS. Win rates are per-series.

config	chain
A	location (EMA) $\to$ leaf
B	location $\to$ scale (EWMA standardize) $\to$ leaf
C	location $\to$ gaussianize $\to$ leaf (raw conformal pattern)
D	location $\to$ scale $\to$ gaussianize $\to$ leaf (normalized pattern)
E	location $\to$ scale $\to$ gaussianize $\to$ AR(1) $\to$ leaf (refit through)
F	location $\to$ scale $\to$ AR(1) $\to$ leaf (same refit, no gaussianize)
G	likelihood-weighted pool over {leaf, A, B, C, D, E, F}
H	likelihood-weighted pool over {leaf, A, B, F} (G minus the gaussianize chains)
I	the pool of G with kernel-smoothed gaussianize

Configs C and D are the conformal production rule written as a chain: transform, apply the fence-post empirical map, stop, and read the predictive off the inverse. Config E continues through the operator. The pools use the same likelihood-weighted online ensembling as the companion paper's forecaster, with a complexity penalty by chain depth. Figure 1 summarizes the comparisons behind the paper's three findings.

4.2 The operator has interior value

In conformal practice the empirical map is terminal. One fits a model, ranks the residuals, and quotes the ranked law as the predictive. The fence-post step is where inference ends, and the design assumes the residuals arrive exchangeable, with nothing left in them worth modeling. Config D is that pattern written as a chain, with one label needing care: every chain ends in a Gaussian leaf with online variance, itself a fitted unconditional model, so D stops at the transform only up to that leaf, and E versus D compares serial conditional refitting with unconditional fitting on the transformed stream. Config E reads the same operator differently: its output is a coordinate system, and whatever structure the model missed is easier to fit in those coordinates. The two readings are testable against each other. For the exact sequential ranks, exchangeable input leaves no serial rank signal for a refit to find, by Proposition 6. The interpolated coordinates carry spacing information besides, and an estimated AR(1) brings estimation noise of its own, so the comparison is informative rather than clean.

On this corpus the second reading wins. Config E improves on config D by 0.053 nats per observation, winning on 75 percent of series, and by 4.0 percent of baseline CRPS, winning on 71 percent. Under the log score the oracle version of this margin is the predictive mutual information remaining in the transformed stream, the quantity the companion paper's ledger prices (Cotton, 2026b). The measured margin is the part of it an AR(1) captures, net of estimation cost. It is evidence that serial structure survives the transform on most of the corpus, and it says the conformal pattern's stopping point is not where the predictive value stops.

A synthetic experiment shows the mechanism in isolation. Let $z_t = 0.8z_{t-1} + \varepsilon_t$ and observe $y_t = \mathrm{sign}(z_t)|z_t|^{1.5}$, a monotone warp of an AR(1), which obscures the linear representation without removing the dependence. Stopping at gaussianize scores -1.56 nats and 0.62 CRPS. Refitting AR(1) through it scores -1.07 and 0.41, beating both the stop and the same AR chain without the operator (-1.16, 0.42). When the downstream learner is well specified only in the empirical map's coordinates, passing through the operator beats both stopping and applying the same learner in the original coordinates.

On the real corpus the comparison with config F, the same refit without the operator, is sharper and score-dependent. Under CRPS the operator helps: E beats F on 65 percent of series, and E is the best single chain in the study at 0.911 of baseline. Under the log score the operator costs 0.108 nats against F and wins only 8 percent of series. We return to this in Section 4.4, because the natural suspicion, that the piecewise-linear density is to blame, turns out to be wrong.

4.3 Nesting limits the cost of admission

The raw conformal pattern C is a reasonable forecaster on this corpus, better than the location baseline on average under both scores. It is also, on a minority of series, a bad choice. Config G, the likelihood-weighted pool over all seven chains, gaussianize chains included, beats the fixed pattern C on 95 percent of series by log score, by 0.249 nats on average. On the twenty series where C is worst, with relative CRPS 1.020, the pool sits at 0.843 on the same series. The pool walks away from the production rule where it underperforms.

Admitting the pattern is approximately free on average. The average hides the shape. Config H, the same pool with the gaussianize chains removed, differs from config G by 0.012 nats and 0.4 percent of baseline CRPS, with the nesting pool G ahead on both means. Per series the picture is lopsided: G wins under a third of series under either score, its median difference from H is zero to four decimals, its fifth-percentile loss is under a thousandth of a nat, and its ninety-fifth-percentile gain is twenty times larger. Frequent negligible losses buy occasional real gains. Under the log score this shape is not luck, and a theorem sits nearby.

Proposition 7 (Nesting is cheap under the log score)**.** *Let the pool predictive be the Bayes mixture over chains with prior π, and let q_t be the predictive of the renormalized sub-mixture over any subset Q of chains, with prior mass π_Q. Then for every realization $y_1, \ldots, y_T$,*

$$-\sum_{t=1}^{T} \log p_t(y_t) \;\le\; -\sum_{t=1}^{T} \log q_t(y_t) \;+\; \log \frac{1}{\pi_Q}.$$

Proof. For a Bayes mixture the cumulative predictive likelihood telescopes to the prior-weighted sum of the chains' likelihoods, $\prod_t p_t(y_t) = \sum_c \pi_c \prod_t p_{t,c}(y_t)$. Dropping the chains outside Q and renormalizing, $\prod_t p_t(y_t) \ge \pi_Q \sum_{c \in Q} (\pi_c/\pi_Q) \prod_t p_{t,c}(y_t) = \pi_Q \prod_t q_t(y_t)$. Take logarithms. $\square$

The total price of admitting the gaussianize chains is at most $\log(1/\pi_Q)$ nats, however the data falls, and the per-observation price vanishes. The proposition applies to the untempered Bayes pool with the depth prior applied once. The implemented pool tempers the likelihood update at rate 0.8 and applies its depth penalty per step, for adaptivity under regime change, so the measured G-versus-H comparison is empirical and does not inherit the exact bound. A runner for the untempered pool ships alongside the benchmark. The measured 0.012 nats in G's favor says this corpus more than repays the admission. No such bound is available to us under CRPS, and there the claim stays empirical. The weights update by likelihood, so the log score does the choosing and CRPS only evaluates the chosen blend. A CRPS-updated pool is the matching experiment, and we leave it open.

This is the practical content of the grammar view. A fixed conformal method is a commitment, and the companion paper prices what the commitment can cost. A grammar that contains the method as one rule converts the commitment into a weighted vote, and the vote is revised online by the scoring rule. Nesting does not avert disasters on this corpus, because the guarded operator does not produce disasters. An exact Bayes pool trims the bad tail at a bounded cumulative price, vanishing per observation. The tempered pool used here exhibits the same behavior empirically.

4.4 Smoothness is not the tax

The log-score cost of the operator invites an implementation explanation: the piecewise-linear empirical map has a piecewise-constant density, and the log score punishes kinks. The kernel-smoothed C^1 variant tests this directly, replacing the interpolated empirical distribution function with a smooth kernel estimate while changing nothing else. The smooth variant tracks the linear one within 0.012 nats and 0.2 percent of baseline CRPS in every position, and against config F it costs 0.113 nats where the linear variant cost 0.108. The tested smoothing does not explain or remove the tax. Alternative bandwidths, adaptive tails, and splines tuned under log loss remain untested, and the remaining evidence points at resolution and tail specification.

The tax is structural, and it is the same structure the companion paper measured for terminal empirical pooling. An empirical map's resolution is bounded by its ranks, and its tails are an assumption, and the log score is exactly the score that pays for resolution and tails. Under CRPS, which pays for calibrated quantiles, the operator earns its place in the chain. Under the log score a fitted parametric leaf on the same stream is worth more. The value of the fence-post operator, like the value of residual pooling in the companion paper, depends jointly on the coordinates, the estimator, and the scoring rule. The grammar does not remove this dependence. It locates it: the scoring rule chooses among chains, and the operator is a candidate, not a default.

The probit is likewise a choice, not the point. Composing the operator with any further fixed invertible link leaves the representation content unchanged, since mutual information is invariant under invertible maps, so the transform-pooling theorem makes the link representation-irrelevant. What remains is model form. On a 120-series holdout check, a logistic link in place of the probit shifts results by hundredths of a nat under both scores, and the shift consistently favors the logistic: the Gaussian leaf pushed through the heavier-tailed link implies a heavier-tailed predictive, which fits real standardized residuals slightly better. The theorem fixes only the representation content. A link can still matter to a restricted downstream learner through specification and estimation, and on this corpus it acts as a small tail-shape choice, an empirical observation rather than a consequence of the theorem.

5 One family

The grammar view collapses a scattered vocabulary. Platt scaling (Platt, 1999), isotonic calibration (Zadrozny and Elkan, 2002), temperature scaling (Guo et al., 2017), quantile recalibration (Kuleshov et al., 2018), conformal predictive systems (Vovk et al., 2019), and distributional conformal prediction (Chernozhukov et al., 2021) can all, in univariate probabilistic forecasting, be represented as monotone recalibration operators on a scalar predictive, score, or probability coordinate, differing in what they fit, what they guarantee, and where in the chain they sit. The forecasting literature's calibration-sharpness tradition (Gneiting et al., 2007; Gneiting and Raftery, 2007) supplies the objective that orders them. In the grammar they are not competing philosophies. They are operators with types, and a pool over chains can hold several at once.

A probabilistic lifting of the same geometry appears in Snell and Griffiths (2025). If $T_{(i)} = F(Y_{(i)})$ are the latent probability levels of the ordered observations, the spacings of the $T_{(i)}$ are Dirichlet$(1, \ldots, 1)$, and the plotting positions $i/(n+1)$ used here are their means. The present operator fixes the mean positions and obtains a deterministic coordinate transform. Snell and Griffiths retain the full spacing distribution and push a loss integral through it, and standard conformal risk control is recovered by replacing the random spacings with their expectations. In grammar terms theirs is a measure-valued operator on the loss coordinate, emitting a distribution over risk bounds rather than a transformed observation, and it is complementary to the observation-coordinate transforms studied here. The two demotions agree: conformal prediction is what remains of an empirical rank geometry after one functional of it is retained and the rest discarded.[2]

Conformal prediction keeps a distinguished place in this family. The fence-post operator is the member whose output carries a finite-sample, distribution-free guarantee at its position in the chain (Proposition 2). No fitted recalibration offers that. The grammar view does not diminish the guarantee. It names what the guarantee attaches to, and it frees the forecaster to keep transforming afterward.

6 Conclusion

Conformal prediction is one production rule in a grammar of composable transforms: model, residual, fence-post empirical map, inverse. Interpolated, the fence-post map is an ordinary invertible online transform, and the rule's distinguishing stop, quoting the empirical law where it stands, becomes a choice the scoring rule can overrule. On 701 economic series, continuing through the operator beats stopping at it on three quarters of the series, a pool that admits the pattern trims its worst cases, at a price bounded for the exact Bayes pool and negligible empirically for the tempered one, and the operator's remaining cost under the log score is unexplained by the tested smoothing and points at resolution and tail specification.

The operator is exceptional as a certificate and ordinary as an architecture, and the certificate it carries is priced in its computational budget: retained-knot resolution and tie mass set the calibration error. The typing question is the theory this view opens. An operator's guarantee attaches to a position in a chain. Which properties survive composition is a discipline we have

[2] One typing caution. The Dirichlet law is a sampling statement about the latent spacings, and reading it as a data-conditional posterior requires an argument about the prior over F that the sampling statement does not supply. Sampling-law, posterior-law, and prequential certificates are different types, and the table should keep them apart.

only begun here, with one operator, a few propositions, and the first rows of a table, and the rest of the table is open.

A Reproducibility

Every number in Section 4 is produced by `benchmarks/gaussianize_chain.py` in the `skaters` repository, and the per-series results are frozen in `benchmarks/gaussianize_chain.csv`. The configuration is as follows. The location operator is an EMA transform with rate 0.1. The scale operator is an EWMA standardization with rate 0.05. The AR(1) operator estimates its coefficient by recursive least squares with forgetting factor 0.99 and ridge 1.0. The gaussianize operator uses at most 39 knots at equal mass from the full sorted history of its inputs (exact order statistics; the buffer is unbounded), a warmup of 30 observations with expanding-moment standardization before it, and a 9-particle equal-probability approximation for the inverse. The kernel-smoothed variant uses a Gaussian-kernel distribution function at Silverman bandwidth, represented by a monotone C^1 cubic with Fritsch–Butland tangents and inverted by Newton's method. The guards are the inverse-slope cap at twenty times the interquartile inverse slope and the downstream-update clamp $|z| \leq 8$.

The pools are tempered likelihood mixtures over the candidate chains of the design table, with tempering rate 0.8, a complexity penalty of 0.005 per unit of chain depth applied to the online weights, and at most 20 mixture components. Scoring starts after a burn-in of 300 observations. The log score mixes each predictive with weight 0.01 toward an expanding-window Gaussian reference and is otherwise exact. CRPS is computed in closed form from the Gaussian-mixture predictive and normalized per series by config A. The corpus is 701 FRED change series with at least 600 observations and repeat fraction below five percent, listed in the frozen results file. Series were downloaded from FRED on 2 August 2026 with observation start 2005-01-01, and non-numeric observations are dropped at parse. The runner does not log guard activations separately. The one recorded activation episode is the warmup event described in Section 2. Sensitivity of the corpus results to the two guard constants has not been measured systematically. The certified propositions do not depend on them, and a sensitivity sweep is future work.

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